Vladimir Bautista Santos

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Professional Summary

Automotive engineer with 16+ years of experience in vehicle development, testing, and validation within OEM and Tier-1 environments. Strong background in Vehicle Attributes including NVH, Thermal, Performance, Durability, and system integration. Experienced in sustaining engineering, root cause analysis, and cross-functional collaboration to support data-driven engineering decisions.

Professional Experience

NVH, Thermal & Performance Test Engineer

Alten USA / Volvo Trucks USA | Jan 2022 – Present

- Product support through the planning, developing, scheduling and execution of project verification and validation engineering activities of varying scale and complexity.
- Provides technical guidance and direction to design engineers and technicians based on verification and validation expertise
- Prepares and communicates testing cost estimations for assigned testing
- Make decisions on types of testing, specifies test procedures, instrumentation and hardware used in the testing of designs.
- Responsible for preparing and communicating conclusions through engineering reports
- Perform system functionality verification testing supporting the following core areas: Noise Vibration and Harshness (NVH), Vehicles
- Dynamics (Handling, Braking, Ride), Dyno Testing (Cooling Performance, Heat Management, HVAC, etc.), and Fuel Economy
- Communication, connectivity and analysis from vehicles using CAN Bus (CAN C, CAN B, CAN FD, Ethernet), LIN Bus networks.

NVH, Thermal & Performance Test Engineer

Stellantis México | Sep 2017 – Dec 2021

- Conducted full-vehicle performance testing, including braking distance, acceleration elasticity, fuel consumption, and gradeability (hill climbing).
- Performed thermal evaluation and validation testing for vehicles sold in the Mexican market, instrumented with 30+ thermocouples covering powertrain, cooling systems, and interior components. Test scenarios included A/C comfort, severe traffic (park idle), and hill climb conditions.
- Executed vehicle communication, connectivity, and data analysis using CAN Bus networks (CAN C, CAN B, CAN FD), Ethernet, and LIN Bus systems.
- Led NVH (Noise, Vibration & Harshness) measurement and analysis activities, establishing and developing NVH test capabilities for Stellantis Mexico, including interior noise measurements and source identification.
- Integrated data acquisition systems by configuring sensors and synchronizing CAN bus signals to correlate noise and vibration sources.
- Performed instrumentation and measurements using strain gauges.
- Led driveability evaluations in Mexico in support of Brazil-based vehicle projects, including test travel support using INCA software.
- Managed material and test equipment procurement for the engineering area.
- Designed, set up, and executed benchmark measurement projects.

Durability & Reliability Test Engineer

Volkswagen México | May 2016 – Aug 2017

- Performed static structural testing on vehicle components, including BIW torsion, bumper deformation, and door hinge evaluations.
- Instrumented components with strain gauges and analyzed deformation and torsional responses.
- Designed and built custom test benches using CATIA V5 to support multi-vehicle testing programs.

NVH Test Engineer

Volkswagen México | May 2013–May 2016

- Led acoustic optimization and NVH validation for special-edition vehicles, evaluating 200+ units annually.
- Conducted dynamic and static NVH testing in compliance with NOM-082-ECOL-1994 using HEAD acoustics and ArtemiS.
- Specialized in acoustic material testing, absorption and transmission loss evaluation, and acoustic package design.
- Designed and executed benchmarking measurements to support competitive analysis.

Mechanical Design Engineer

Kobolds Technologies — May 2011–May 2016

- Developed 2D and 3D CAD models for industrial equipment modifications.
- Gained international experience in PLM systems and Solidworks during a residency at GEA Group (Netherlands).
- Led a four-engineer team in a CAD conversion project for ASSA ABLOY NEMEF.

Maintenance Engineer

Soporte CMM — Sep 2009–Mar 2011

- Performed repair, instrumentation, calibration, and preventive maintenance of CMMs (Coordinate Measuring Machines).
- Retrofitted CMMs by upgrading hardware and software components across various machine types.
- Provided service and technical support to industrial clients, including Nemak and Meggitt.

Education

Master in Data Science (In Progress – Aug 2024)
Universidad del Valle de México

Master in Business Administration (Mar 2012 – Mar 2014)
Benemerita Universidad Autonoma de Puebla (coursework completed)

B.S. Mechatronic Engineering
Universidad Popular Autonoma del Estado de Puebla

Certifications

Machine shop & Software Developer

Core Skills

Product Validation & Testing · Design Failure Mode and Effects Analysis (DFMEA) · Root Cause Analysis · Design Verification Plan & Report (DVP&R) · Cross-functional Collaboration · Data-driven Decision Making · Timing & Schedule Control · Continuous Improvement · Technical Decision-Making · Data Analysis & Signal Processing · CAN, LIN, CAN FD & Ethernet Networks · Sustaining Engineering & Change Management · Vehicle Attribute Target Setting (NVH, Thermal, Ride & Handling, Performance) · Benchmarking & Competitive Analysis.

Tools & Software

CATIA · Solidworks · AutoCAD · Power BI · Simcenter Testlab · PROTOM (Build of Materials) · CANalyzer / CANoe · Ipemotion · ArtemiS · Dewesoft · MS Project · Python · SQL · PowerApps